Young's Modulus

How stiff is the *stuff*, as opposed to how stiff is the *thing*

Hang a mass from a steel cable and the cable stretches — a little. Let go and it springs back. So a cable is a spring, and it must have a spring constant k. But k depends on how long the cable is and how thick it is, which are facts about *that particular cable*, not about steel. Young's modulus is the number that describes the steel itself. Everything in this handout follows from separating those two ideas. There is a total of **44** points; problems marked **[A]** are a little harder.

1 Stress, Strain, and the Definition

Idea 1: Two ratios, and the number connecting them

Take a rod or wire of length ℓ and cross-sectional area A, and pull on it with force F along its length. Define

$$\textbf{stress} \quad \sigma = \frac{F}{A}, \qquad\qquad \textbf{strain} \quad \varepsilon = \frac{\Delta\ell}{\ell}.$$

Stress is "force spread over the area that carries it." Strain is "fractional stretch" — how much longer, as a fraction of what you started with. For small stretches the two are proportional, and the constant of proportionality is **Young's modulus** Y:

$$\boxed{\sigma = Y\varepsilon} \qquad \text{that is,} \qquad \boxed{\frac{F}{A} = Y\,\frac{\Delta\ell}{\ell}}$$

This is Hooke's law, rewritten so that only the *material* appears. Y is a property of steel, or copper, or bone — not of the particular piece you happen to be holding.

Remark

Why divide by A? Because a rope twice as thick shares the load between twice as much material, so each part of the material is pulled half as hard. Why divide by ℓ? Because a rope twice as long has twice as much material to stretch, so the same *fractional* stretch produces twice the absolute stretch. Stress and strain are built precisely to cancel out the size of the object.

[3] **Problem 1.** A wire of cross-sectional area $A = 2.0\ \text{mm}^2$ carries a tension of $F = 400$ N.

(a) What is the stress, in pascals?

(b) If the wire is 1.5 m long and stretches by 0.60 mm, what is the strain?

2 Units, and Why They Are the Units of Pressure

Idea 2: Y is measured in pascals

Strain is a length divided by a length, so it is **dimensionless**. That means $Y = \sigma/\varepsilon$ has exactly the same units as stress:

$$[Y] = \frac{[F]}{[A]} = \frac{\mathrm{N}}{\mathrm{m}^2} = \mathrm{Pa}, \qquad \text{and in base units} \qquad [Y] = \frac{MLT^{-2}}{L^2} = ML^{-1}T^{-2}.$$

Those are the dimensions of pressure, and of energy per unit volume — all three are the same thing dimensionally. Real values are large, so Y is usually quoted in gigapascals (1 GPa = 10^9 Pa):

rubber	0.01–0.1 GPa	aluminum	70 GPa
nylon	3 GPa	steel	200 GPa
wood (along the grain)	10 GPa	tungsten	400 GPa
bone	15 GPa	diamond	1000 GPa

[2] **Problem 2.** Show, using base dimensions M, L, T, that Young's modulus has the same dimensions as energy per unit volume. (This is not a coincidence: stretching a wire stores elastic energy, and Y sets how much.)

Idea 3: An intuition for why the number is so big

Put $\varepsilon = 1$ into $\sigma = Y\varepsilon$. You get $\sigma = Y$. So *Young's modulus is the stress you would need to double the length of the object* — if the material kept behaving linearly all the way there, which of course it never does. Steel snaps at a stress of roughly 0.4 GPa, some 500 times smaller than its $Y = 200$ GPa. Turned around, that says steel breaks at a strain of about $1/500 = 0.2\%$. **Real strains in stiff materials are tiny**, which is why the modulus is huge.

[3] **Problem 3.** Nylon has $Y \approx 3$ GPa and breaks at a stress of about 75 MPa. At what strain does it break, and how does that compare with steel's 0.2%? Does the more easily broken material stretch more or less before failing?

3 From Young's Modulus to a Spring Constant

Idea 4: $k = YA/\ell$ — the most useful line in this handout

Rearrange $F/A = Y\Delta\ell/\ell$ to put it in the form $F = k\,\Delta\ell$:

$$F = \frac{YA}{\ell}\,\Delta\ell \qquad \Longrightarrow \qquad \boxed{k = \frac{YA}{\ell}}$$

Read this carefully, because it is the whole point:

- Y belongs to the **material**. Cut a steel cable in half and both halves still have $Y =$ 200 GPa.

- k belongs to the **object**. Cut the cable in half and each half has *twice* the spring constant of the original.
- **Longer $\Rightarrow$ floppier.** $k \propto 1/\ell$.
- **Thicker $\Rightarrow$ stiffer.** $k \propto A$, so $k \propto r^2$ for a circular wire of radius r.

Remark

You can get both dependences without any algebra, just from how springs combine. A wire of length 2ℓ is two wires of length ℓ joined end to end — springs in **series**, and $1/k_{\text{tot}} = 1/k + 1/k$ gives $k_{\text{tot}} = k/2$. A wire of area $2A$ is two wires of area A lying side by side sharing the load — springs in **parallel**, and $k_{\text{tot}} = k + k = 2k$. Both match $k = YA/\ell$ exactly.

[4] **Problem 4.** A steel wire ($Y = 200$ GPa) has length $\ell = 2.0$ m and cross-sectional area $A = 1.0\ \text{mm}^2$. A 20 kg mass hangs from it. Take $g = 10\ \text{m/s}^2$.

(a) Find the spring constant k of the wire.

(b) Find how far it stretches.

[3] **Problem 5.** Two wires are made of the same metal. Wire 2 is three times as long as wire 1 and has twice the radius. By what factor is wire 2's spring constant larger or smaller than wire 1's? By what factor does their Young's modulus differ?

[3] **Problem 6.** A rubber band and a steel wire of identical dimensions are each stretched by the same *force*. Which stretches further, and by roughly what factor? (Use $Y_{\text{rubber}} \approx 0.05$ GPa.)

4 Oscillations on a Wire

Idea 5: A hanging mass on a wire is just a mass on a spring

Once you know $k = YA/\ell$, vertical oscillations need nothing new:

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{YA}{\ell m}}.$$

Gravity does not appear. It shifts the equilibrium point down by mg/k but does not change the stiffness, exactly as for an ordinary vertical spring.

Example 1: The question you saw on the practice exam

Steel has $Y = 200$ GPa. A mass $m = 300$ kg hangs from a steel wire of circular cross-section with radius $r = 0.15$ cm and length $\ell = 2$ m. Find the frequency of vertical oscillations.
First the area: $r = 0.15\ \text{cm} = 1.5 \times 10^{-3}$ m, so

$$A = \pi r^2 = \pi(1.5 \times 10^{-3})^2 = 7.07 \times 10^{-6}\ \text{m}^2.$$

Then the spring constant:

$$k = \frac{YA}{\ell} = \frac{(200 \times 10^9)(7.07 \times 10^{-6})}{2} = 7.07 \times 10^5 \text{ N/m}.$$

Then the frequency:

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{7.07 \times 10^5}{300}} = \frac{1}{2\pi}\sqrt{2360} = \frac{48.6}{2\pi} \approx 7.7 \text{ Hz}.$$

The only step that is specific to this topic is $k = YA/\ell$. Everything after it is ordinary SHM. Watch the centimetres: writing $r = 0.15$ instead of 0.0015 costs a factor of 100 in r, hence 10^4 in A and 100 in f.

[3] **Problem 7.** In the example above, how far below the wire's unstretched end does the mass hang at equilibrium? Compare that sag with the amplitude of a small oscillation, and comment on whether the wire ever goes slack.

5 Dimensional Analysis with Y

Idea 6: $\sqrt{Y/\rho}$ is a speed

Y has dimensions $ML^{-1}T^{-2}$ and density ρ has dimensions ML^{-3}. Divide:

$$\left[\frac{Y}{\rho}\right] = \frac{ML^{-1}T^{-2}}{ML^{-3}} = L^2T^{-2},$$

which is a velocity squared. So $\sqrt{Y/\rho}$ is the only speed you can build from a material's stiffness and its density — and it is, up to a factor of order 1, the **speed of sound in that material**. Stiffer material, faster sound; heavier material, slower sound.

[3] **Problem 8.** Estimate the speed of sound in steel, using $Y = 200$ GPa and $\rho = 8000$ kg/m^3. Compare with the speed of sound in air (340 m/s).

[4] **Problem 9.** [A] A wire of length ℓ, area A, density ρ and Young's modulus Y hangs with a mass m on the end. Without solving anything, use dimensional analysis to argue that the oscillation frequency must have the form $f = \frac{1}{2\pi}\sqrt{\frac{YA}{\ell m}} \times g(\text{dimensionless})$, and identify a dimensionless ratio the leftover function could depend on.

6 Common Traps

Remark

- **"Longer means less stiff" applies to k, never to Y.** A longer rope has a smaller spring constant and the identical Young's modulus. If a problem changes the length and you find yourself changing Y, stop.

- **Watch the area units.** 1 $\text{mm}^2 = 10^{-6}\ \text{m}^2$ and 1 $\text{cm}^2 = 10^{-4}\ \text{m}^2$. Squaring turns a small slip into a factor of 100 or 10^4.
- **A radius given in centimetres must be converted before squaring.** This is the single most common way to lose the steel-wire problem.
- **Strain is a fraction, not a length.** $\Delta\ell$ is a length; $\Delta\ell/\ell$ is the strain. Only the second one goes into $\sigma = Y\varepsilon$.
- **Stiff is not the same as strong.** Y tells you how much it stretches under load; the breaking stress tells you when it snaps. Materials can be high in one and low in the other.

7 F = ma Style Questions

Four questions in the style of the exam. Answer each in about two minutes.

[4] **Problem 10.** A steel wire of length ℓ and cross-sectional area A stretches by 1.0 mm when a certain mass is hung from it. A second wire, made of the same steel with the same cross-sectional area but *twice the length*, carries the same mass. Compared with the first wire, the second wire has

(A) the same Young's modulus, half the spring constant, and stretches 2.0 mm.
(B) half the Young's modulus, half the spring constant, and stretches 2.0 mm.
(C) the same Young's modulus, twice the spring constant, and stretches 0.50 mm.
(D) the same Young's modulus, the same spring constant, and stretches 1.0 mm.
(E) twice the Young's modulus, the same spring constant, and stretches 0.50 mm.

[4] **Problem 11.** A steel wire ($Y = 200$ GPa) of length 2.0 m and cross-sectional area 1.0 mm^2 supports a 20 kg mass. Taking $g = 10\ \text{m/s}^2$, the wire stretches by about

(A) 0.20 mm
(B) 0.50 mm
(C) 1.0 mm
(D) 2.0 mm
(E) 4.0 mm

[4] **Problem 12.** The speed v of a sound pulse traveling along a thin solid rod depends only on the rod's Young's modulus Y and its density ρ. For steel, $Y = 200$ GPa and $\rho = 8000\ \text{kg/m}^3$. The speed is closest to

(A) 160 m/s
(B) 1600 m/s
(C) 5000 m/s
(D) 25000 m/s
(E) 2.5×10^7 m/s

[4] **Problem 13.** A mass m hangs from a steel wire of length ℓ and cross-sectional area A, and oscillates vertically with frequency f. The wire is replaced by one of the same steel and the same length, but with *four times* the cross-sectional area. The new frequency is

(A) $f/4$
(B) $f/2$
(C) f
(D) $2f$
(E) $4f$

End of handout. If you remember only one line, make it $k = YA/\ell$: the material contributes Y, the object contributes A and ℓ, and everything else is the spring physics you already know.

Solutions

Problem 1. (a) $A = 2.0 \text{ mm}^2 = 2.0 \times 10^{-6} \text{ m}^2$, so $\sigma = F/A = 400/(2.0 \times 10^{-6}) = 2.0 \times 10^8$ Pa $=$ 200 MPa.

(b) $\varepsilon = \Delta\ell/\ell = (0.60 \times 10^{-3})/1.5 = 4.0 \times 10^{-4}$. Note this is a pure number — no units.

Problem 2. Energy has dimensions ML^2T^{-2} and volume has dimensions L^3, so energy per volume is $ML^2T^{-2}/L^3 = ML^{-1}T^{-2}$. That is exactly the result computed above for Y.

Problem 3. $\varepsilon = \sigma/Y = (75 \times 10^6)/(3 \times 10^9) = 0.025$, i.e. 2.5% — about twelve times the strain steel tolerates. Nylon fails at a much *lower stress* but a much *higher strain*. "Strong" (large breaking stress) and "stiff" (large Y) are different properties, and nylon is neither as strong nor as stiff as steel, yet it stretches far more before it goes.

Problem 4. (a) $k = YA/\ell = (200 \times 10^9)(1.0 \times 10^{-6})/2.0 = 1.0 \times 10^5$ N/m.

(b) The tension is $F = mg = 200$ N, so $\Delta\ell = F/k = 200/(1.0 \times 10^5) = 2.0 \times 10^{-3}$ m $= 2.0$ mm.

As a check, the strain is 2.0 mm/2.0 m $= 10^{-3}$, and the stress is 200 N$/10^{-6}$ m$^2 = 2 \times 10^8$ Pa; their ratio is 2×10^{11} Pa $= Y$, as it must be.

Problem 5. $k = YA/\ell \propto r^2/\ell$. Doubling r multiplies k by 4; tripling ℓ divides it by 3. So $k_2/k_1 = 4/3$. Young's modulus is identical for both — they are the same metal — so it differs by a factor of 1.

Problem 6. Same F, same A means same stress, so $\varepsilon = \sigma/Y$ says the strain is inversely proportional to Y. The rubber stretches by a factor $Y_{\text{steel}}/Y_{\text{rubber}} = 200/0.05 = 4000$ more than the steel.

Problem 7. The static stretch is $\Delta\ell = mg/k = (300)(9.8)/(7.07 \times 10^5) = 4.2 \times 10^{-3}$ m ≈ 4.2 mm. Any oscillation with amplitude smaller than 4.2 mm keeps the wire in tension the whole time, so the motion stays simple harmonic. A larger amplitude would let the wire go slack at the top of the motion, where it can push nothing — and then the motion is no longer SHM.

Problem 8. $v = \sqrt{Y/\rho} = \sqrt{(2 \times 10^{11})/(8 \times 10^3)} = \sqrt{2.5 \times 10^7} = 5.0 \times 10^3$ m/s. That is about 15 times the speed of sound in air. (The measured value for steel is about 5900 m/s, so the estimate is good.)

Problem 9. YA has dimensions $(ML^{-1}T^{-2})(L^2) = MLT^{-2}$, a force. Dividing by ℓm gives $MLT^{-2}/(LM) = T^{-2}$, so $\sqrt{YA/(\ell m)}$ is a frequency — the only one you can build from those four quantities. Anything else must be dimensionless. The natural dimensionless ratio available is the mass of the wire to the hanging mass, $\rho A\ell/m$; the exact answer does depend on it slightly, because the wire itself has to move too. When the wire is light compared with the load, that ratio is near zero and $g \to 1$, recovering the formula used above.

Problem 10. **(A).** Young's modulus is a property of steel and does not know how long the wire is, so it is unchanged. The spring constant $k = YA/\ell$ is halved when ℓ doubles. With the same force and half the spring constant, $\Delta\ell = F/k$ doubles to 2.0 mm.

Equivalently and even faster: the same mass on the same cross-section gives the same stress, hence the same *strain*. Same fractional stretch on twice the length is twice the absolute stretch.

Problem 11. **(D).** The tension is $F = mg = 200$ N, so the stress is $\sigma = 200/(1.0 \times 10^{-6}) = 2.0 \times 10^8$ Pa. The strain is $\varepsilon = \sigma/Y = (2.0 \times 10^8)/(2.0 \times 10^{11}) = 1.0 \times 10^{-3}$. Therefore $\Delta\ell = \varepsilon\ell = (1.0 \times 10^{-3})(2.0 \text{ m}) = 2.0 \times 10^{-3}$ m $= 2.0$ mm.

Problem 12. **(C).** Only one combination of Y and ρ has dimensions of speed. Since $[Y] = ML^{-1}T^{-2}$ and $[\rho] = ML^{-3}$, the ratio Y/ρ has dimensions L^2T^{-2}, so $v = \sqrt{Y/\rho}$. Numerically $v = \sqrt{(2 \times 10^{11})/(8 \times 10^3)} = \sqrt{2.5 \times 10^7} = 5.0 \times 10^3$ m/s.

Distractor (E) is Y/ρ without the square root, and (A) is $\sqrt{\rho/Y}$ scaled — both are what you get by taking the combination in the wrong order, which the dimensions rule out immediately.

Problem 13. **(D).** The spring constant is $k = YA/\ell$, so quadrupling A quadruples k. Since $f = \frac{1}{2\pi}\sqrt{k/m}$ and m is unchanged, $f \propto \sqrt{k}$, and $\sqrt{4} = 2$. The frequency doubles.

Note that Y never changes — it is the same steel throughout. Only the geometry moved.